



Estimation of the two-stanza growth curves with age determination uncertainty for Pacific blue marlin (*Makaira nigricans*) in the western and central North Pacific Ocean

Yi-Jay Chang ^{1,*}, Tamaki Shimose², Xu-Bang Chang¹, Ayumu Furuyama³, Jhen Hsu¹, Minoru Kanaiwa³, Wei-Chuan Chiang ⁴, Jon K.T. Brodziak⁵, Mikihiro Kai ⁶, Hirotaka Ijima⁶, Michelle L. Sculley ⁵, Michael J. Kinney⁵

¹Institute of Oceanography, National Taiwan University, Taipei, 10617, Taiwan

²Sanriku Fisheries Research Center, Iwate University, Iwate, 020-8550, Japan

³Mie University Graduate School, Faculty of Bioresources, Tsu, 514-8507, Japan

⁴Eastern Fishery Research Center, Fisheries Research Institute, Taitung, 96143, Taiwan

⁵NOAA National Marine Fisheries Service, Pacific Islands Fisheries Science Center, Honolulu, 96818, United States

⁶Fisheries Resources Institutes, Japan Fisheries Research and Education Agency, Yokohama, 236-8648, Japan

*Corresponding author. Institute of Oceanography, National Taiwan University, Taipei 10617, Taiwan. E-mail: yjchang@ntu.edu.tw

Abstract

The Pacific blue marlin (*Makaira nigricans*) is a highly migratory apex predator of ecological and economic importance. However, uncertainties in age and growth estimates remain a challenge for stock assessments. This study aims to refine these estimates by integrating otolith micro-increment counts from 13 juveniles (115–152 cm eye-to-fork length, EFL) and sectioned dorsal fin spines from 504 juveniles and adults (115–269.4 cm EFL) collected off Taiwan waters, together with historical length-at-age data compiled by the ISC International Billfish Biological Sampling Program using fin spine- ($n = 591$, 118–321.9 cm EFL) and otolith-based ($n = 27$, 17.2–159.7 cm EFL) methods and bomb radiocarbon dating ($n = 1$, 330.5 cm EFL) on specimens sampled across the western and central North Pacific Ocean (WCNPO). The combined otolith dataset ($n = 40$) yielded daily age estimates ranging from 26 to 349 days, with the Gompertz model predicting an EFL of 146.92 cm at age one. Port-sampled blue marlin from Taiwan showed similar observed age ranges for both sexes: females aged 1.5–11 years (142.8–269.4 cm EFL) and males 1–10.5 years (133–208 cm EFL). Across the WCNPO, the combined spine and bomb radiocarbon dataset showed age ranges of 1–27 years for females (136–330.5 cm EFL) and 1–18 years for males (118–232.1 cm EFL). Growth was sexually dimorphic, with females growing faster and consistently attaining greater mean lengths-at-age than males from age one onward. Mean marginal increment ratio analysis indicated annual growth band formation primarily in Sep–Oct. Sex-specific von Bertalanffy and two-stanza growth models (TSGM) were fitted using a Bayesian errors-in-variables approach to account for age determination uncertainty. The TSGM better described growth patterns, capturing a transition near age one and providing an improved fit, particularly for older individuals, as indicated by lower Deviance Information Criterion values. This is the first study on blue marlin growth to cover a broader size range and spatial distribution in the WCNPO, enhancing the accuracy of growth parameter estimates for stock assessment models. The study also addresses limitations and outlines potential directions for future research.

Keywords: billfish; age determination error; otolith; fin spines; growth modeling; Pacific Ocean

Introduction

The blue marlin (*Makaira nigricans*) is a large apex predatory billfish distributed circumtropically throughout the world's tropical and subtropical marine waters (Nakamura 1985, Cyr et al. 1990, Prince et al. 1991). In the Pacific, bycatch from various commercial longline fleets targeting tuna—such as those from Japan, Taiwan, the USA, and other members of western and central Pacific Fisheries Commission (WCPFC)—have historically been the primary source of fishing mortality for blue marlin. Blue marlin can cover vast distances, as shown by transoceanic, transequatorial, and interoceanic movements identified in mark-recapture studies (Orbesen et al. 2008). Genetic studies (Buonaccorsi et al. 2001, Graves and McDowell 2003, Sorenson et al. 2013, Williams et al. 2020) and fishery data (Hinton 2001, Kleiber et al. 2003) suggest that there is a single stock of blue marlin in the Pacific Ocean. Catches of Pacific blue marlin over the past decade have remained sta-

ble at ~20 000 metric tons. The most recent stock assessment conducted by International Scientific Committee for Tuna and Tuna-like Species in the North Pacific (ISC 2021) determined that the stock is not overfished and not experiencing overfishing relative to maximum sustainable yield-based reference points, although there is considerable uncertainty in recent input catch-per-unit-effort data and life-history parameters.

Advanced methodologies employed in integrated stock assessments, such as Stock Synthesis (Methot and Wetzel 2013), rely heavily on accurate estimates of age and growth rates (Maunder and Punt 2013, Francis 2016). However, there is a notable lack of reliable age and growth information for Pacific blue marlin, which has been identified as a significant source of uncertainty in the ISC stock assessment reports (ISC 2016, 2021). Research on the age and growth of the Pacific blue marlin has been relatively limited. Hill et al. (1989) evaluated four different hard parts (sectioned dorsal fin spines; anal fin

spines; vertebral bones; and otolith surface ridges) for age determination of the Pacific blue marlin caught around Hawaii. Wilson *et al.* (1991) used the number of otolith surface ridges to estimate the age composition of the Pacific blue marlin. Shimose *et al.* (2015) used the combined method of sagittal otolith micro-increment counts of juvenile individuals (0–1 ages) and sectioned dorsal fin spines of subadult and adult individuals (1–19.5 ages) for age determination and growth estimation. Furthermore, otoliths from a “grander” female Pacific blue marlin (330.5 cm eye-to-fork length [EFL]) were analyzed using bomb radiocarbon (^{14}C) dating by Andrews *et al.* (2018), which estimated a longevity of ~20 years. This estimate was based on multiple lines of evidence, including radiocarbon levels in the sagitta core and whole lapillus, as well as the expected geographic distribution and bomb radiocarbon exposure range encountered by the species.

Despite considerable efforts in previous studies, significant challenges remain in understanding the age and growth dynamics of Pacific blue marlin. First, much of the existing research has relied on regional, fishery-dependent samples, which are often constrained by the inability to capture data across the species' full spatial distribution and size range. These limitations arise from restrictions in the operational range of fishing vessels and the selection of gear types. Second, the vascularization of the spine core and the presence of false annual growth bands are common in billfish species, especially for larger/older individuals (Drew *et al.* 2006, Kopf *et al.* 2011, Shimose *et al.* 2015), complicating the interpretation of growth rings and leading to substantial errors in age determination. Third, in highly migratory species, multiple distinct growth phases can often be identified throughout the life cycle (Eveson *et al.* 2015). Recent studies on the growth patterns of such species have suggested that a two-stanza growth model (TSGM) is more suitable than the von Bertalanffy growth model (VBGM) in accounting for changes in growth during the transition from juvenile to subadult/adult stages (Hearn and Polacheck 2003, Dortel *et al.* 2013).

To address the challenges and limitations in previous studies on the age and growth of Pacific blue marlin, the objectives of this study are as follows: (1) to incorporate dorsal fin spine section samples collected from waters off Taiwan, in combination with historical length-at-age data compiled through the ISC International Billfish Biological Sampling (IBBS) Program, which includes specimens from the western and central North Pacific Ocean (WCNPO), in order to encompass a broader size range and spatial distribution for the species; (2) to enhance age determination by employing a combined method of dorsal fin spine sections and sagittal otolith micro-increments (used as a validation tool) to reduce age-reading errors; (3) to apply a TSGM that accounts for changes in growth rates across different life stages, while incorporating a Bayesian errors-in-variables (EIV) approach (i.e. measurement-error modeling) to represent the sources of uncertainty associated with age estimation, thereby improving the estimation of growth curves and parameter estimates.

Materials and methods

Sample collection from Taiwan waters

Data on EFL (measured to the nearest cm), round weight (RW) (whole fish measured to the nearest 0.1 kg), and dorsal fin spines from 616 Pacific blue marlin individuals caught

in the waters off Taiwan were collected monthly at the fishing ports of Tungkang, Shinkang, and Nanfungao from Sep 2016 to Jan 2018. Among the collected specimens, 50 pairs of sagittal otoliths from juveniles with matching sets of dorsal fins (spines), collected between Nov 2016 and Apr 2017, were selected for otolith micro-increment analysis. Sex determination was carried out through the examination of macroscopic and microscopic gonadal characteristics, following the methods outlined by Merrett (1970, 1971), Sun *et al.* (2009), and Kopf *et al.* (2012). Preparing sagittal otoliths from juvenile blue marlin was technically difficult due to the small size and fragility of the structures. After excluding otoliths that were damaged during extraction, fractured during transverse cutting, or over-polished during otolith section preparation, 25 male sagittal otolith samples (115–172 cm EFL) out of the original 50 samples were successfully retained for daily growth ring analysis. Additionally, after excluding samples with unclear or irregular growth rings, as well as those exhibiting such features beyond the vascularized region, a total of 504 spine sections were successfully used for annual age determination. Among these, 277 spine sections were from females (EFL range: 142.8 cm to 269.4 cm), and 227 were from males (EFL range: 115 cm to 208 cm).

Data from the ISC IBBS Program

Additional data were collected from the ISC International Billfish Biological Sampling (IBBS) Program (Kinney and O'Malley 2020), which is a collaborative data-sharing and biological sampling effort involving Japan, Taiwan, and the USA. The program aims to improve estimates of growth, maturity, and stock structure by developing a uniform sampling protocol applicable across the North Pacific (Kinney *et al.* 2023). The present study utilizes historical length-at-age data for Pacific blue marlin compiled through the IBBS Program, which includes specimens from the WCNPO. For consistency, all lengths reported in this section were converted to EFL, with conversion details provided in [Supplementary Table S1](#). The IBBS datasets incorporated into this analysis include the following sources: (1) Shimose *et al.* (2015)—This dataset includes juvenile size measurements and age estimates based on the enumeration of otolith daily growth micro-increments, with 27 individuals ranging from 17.2 to 159.7 cm EFL, sampled across the WCNPO. Sampling covered a latitudinal range of 4.3°S to 26.5°N and a longitudinal range of 122.95°E to 135.58°W. The dataset also includes adult size and age data based on counts of annual growth bands in dorsal spine sections collected near Yonaguni Island, Japan. Two older females were included in the present study (284.1 cm EFL at age 13.5; 270.3 cm EFL at age 19.5), as samples from older individuals are of limited availability and provide critical information for characterizing the growth curve. (2) Andrews *et al.* (2018)—This study validates an adult longevity estimate of ~20 years using on bomb radiocarbon dating of otoliths from an adult female (330.5 cm EFL) sampled off Hawaii, USA. (3) Hill *et al.* (1989)—Thirteen older individuals, aged using annual growth bands in anal fin spines and also collected from Hawaiian waters, were included in the present study. These consisted of seven females (244.7–321.9 cm EFL, ages 16–27) and six males (174.7–232.1 cm EFL, ages 12–18). Together with the two older females from Shimose *et al.* (2015), these individuals enhanced representation of large and older size classes in the present analysis. (4) Japan

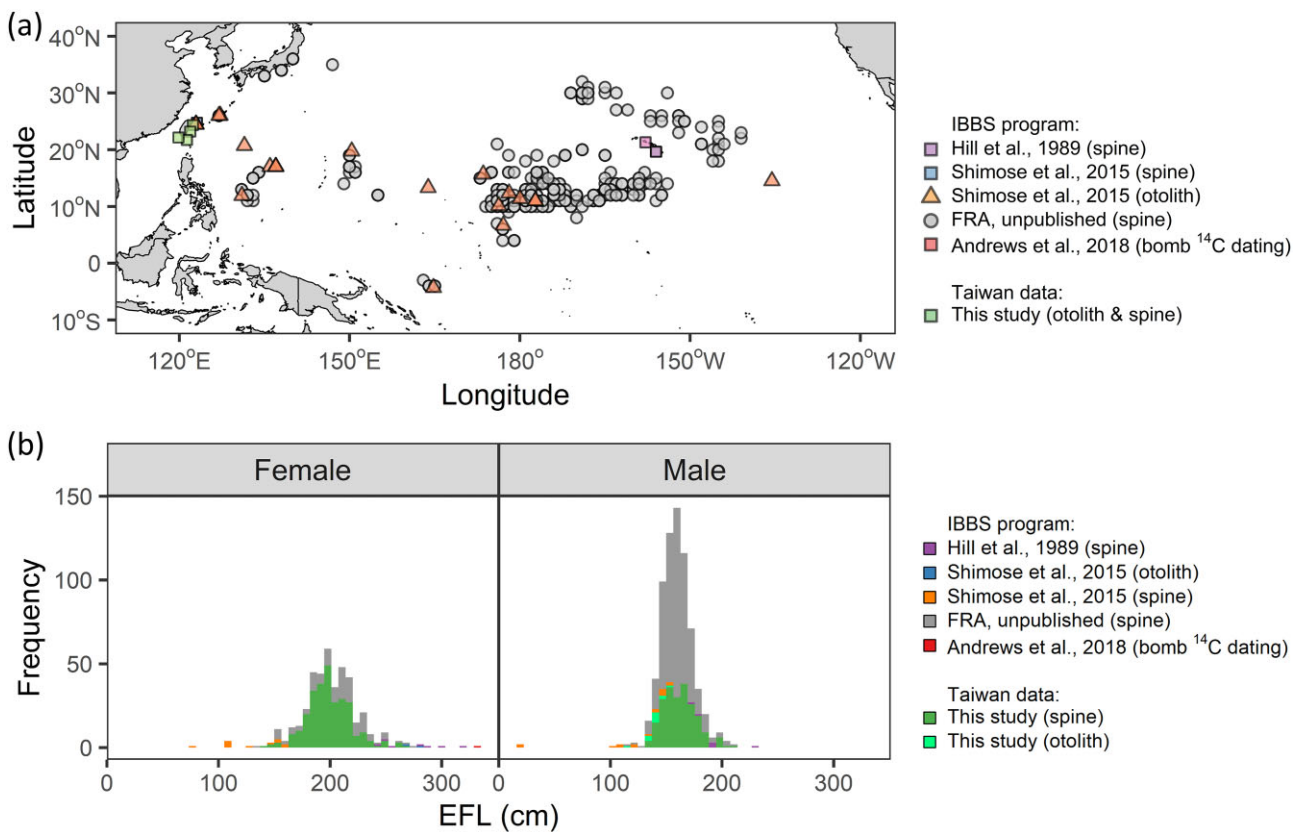


Figure 1. (a) Map of sampling locations for the Pacific blue marlin (*Makaira nigricans*) used in the age and growth study, including samples collected in the waters off Taiwan and length-at-age data from the ISC IBBS Program (specimens from the western and central North Pacific Ocean); (b) Eye-to-fork length (EFL) frequency distribution of the female (left panel) and male (right panel) specimens used in the study. Length-at-age data from the ISC IBBS Program includes estimates of unpublished spine annual age estimation data from the Fisheries Research Agency (FRA, unpublished data), spine annual age estimation of Hill et al. (1989), otolith daily age estimation and spine annual age estimation of Shimose et al. (2015), and otolith bomb radiocarbon dating of Andrews et al. (2018). Four otolith samples of unknown-sex individuals from Shimose et al. (2015) were grouped under the male category in the figure.

Fisheries Research and Education Agency (FRA, unpublished)—This dataset, extracted from the IBBS database and following the age estimation protocol established by Kopf et al. (2011) and Shimose et al. (2015). Missing growth rings were estimated using the replacement method of Hill et al. (1989) and Chiang et al. (2004). The dataset provides spine-based age information for specimens collected from the WCNPO. The dataset includes 121 females (136–275 cm EFL, ages 1–8.5) and 453 males (118–212 cm EFL, ages 1–5), sampled across 4.3°S–36.5°N and 127°E–141°W. Detailed information on sampling region, sampling year, size range, and age estimation method of the length-at-age data of the Pacific blue marlin compiled in this study are shown in Fig. 1 and Table 1. Collectively, these samples cover nearly the full size range of harvested individuals and offer broader spatial coverage across the WCNPO compared to previous studies.

Otolith analysis of juvenile specimens from waters off Taiwan

The otoliths of juvenile specimens from the waters off Taiwan were embedded in resin and transversely sectioned to 1-mm thick before being attached to glass slides with reference to Shimose et al. (2015). Mounted sections were examined under a light microscope using transmitted light, pol-

ished sequentially with 2400–4000 grit wet/dry sandpapers, then with 3 μm aluminum powder until the core was exposed, and finally with 0.05 μm aluminum powder. To enhance growth ring visibility, 0.01 N EDTA and 5% HCl were applied to the otoliths. Photographs were taken with a Leica 020–518.500 DM/LS light microscope at $\times 200$ –400 magnification to discern micro-increments from the primordium core to the edge of the ventral lobe (Kopf et al. 2011, Shimose et al. 2015). Primary increments, as detailed by Prince et al. (1991), were counted twice using no information on the individual specimen. It was assumed that otolith micro-increments formed daily, similar to growth studies for blue marlin (Prince et al. 1991) and striped marlin (Kopf et al. 2011). Based on the exploratory analysis and the results from Shimose et al. (2015), after the ventral arm inflection in otolith sections, micro-increments became very small and discontinuous at ~ 1 year of age. Therefore, only otolith samples with daily age counts of less than 365 days were used in further analyses. The back-calculated hatch dates were then compared with the documented spawning months of May to Sep for blue marlin in the western North Pacific (Shimose et al. 2009, Sun et al. 2009). The precision of micro-increment counts was assessed using Average Percent Error (APE; Beamish and Fournier 1981) and the Coefficient of Variation (CV) (Campana 2001).

Table 1. Summary of Pacific blue marlin (*Makaira nigricans*) specimens used in this study for age and growth analysis. The dataset includes otolith- and spine-based length-at-age information derived from both the ISC International Billfish Biological Sampling (IBBS) Program (covering the western and central North Pacific Ocean) and samples collected from waters off Taiwan. All lengths reported in this table were converted to eye-to-fork length (EFL), with conversion details provided in [Supplementary Table S1](#). Geographic coverage and sampling port coordinates are provided in the footnotes.

Data source	Region	Sampling year	Sample size	Range of size (cm EFL)	Age estimation method	Reference
IBBS Program	Waters off Hawaii, USA	1982–1983	13	174.7–321.9	Spine	Hill et al. (1989)
	Waters off Yonaguni Island, Japan	2003–2006	2	270.3; 284.1	Spine	Shimose et al. (2015)
	Western and Central North Pacific Ocean ^a	2004–2008	27	17.2–159.7	Otolith	Shimose et al. (2015)
	Western and central North Pacific Ocean ^b	2003–2008	574	118–275	Spine	FRA (<i>unpublished</i>)
	Waters off Hawaii, USA	2009	1	330.5	Otolith bomb radiocarbon dating	Andrews et al. (2018)
Taiwan data	Waters off Taiwan ^c	2016–2018	25	115–172	Otolith	This study
			504	115–269.4	Spine	

^aThe geographic range of the samples spans from 4.3°S to 26.5°N in latitude, and from 122.95°E to 135.58°W in longitude.

^bThe geographic range of the samples spans from 4.3°S to 36.5°N in latitude, and from 127°E to 141°W in longitude.

^cPort sampling was conducted at the fishing ports of Tungkan (22.45°N, 120.43°E), Shinkang (23.09°N, 121.38°E), and Nanfangao (24.58°N, 121.87°E), Taiwan.

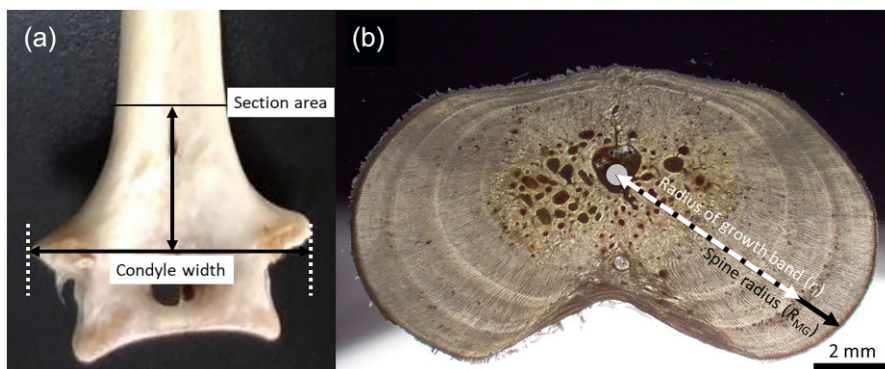


Figure 2. Photographs of (a) the fifth spine from the first dorsal fin of a Pacific blue marlin (*Makaira nigricans*) with an eye-to-fork length of 175 cm, illustrating the transverse sectioning position at half the condyle width for age estimation; and (b) an explanation of the measurements of the spine radius (R_{MG}) and radius of growth band (r_i) within the spine section, viewed under transmitted light. The gray circle represents the core (focus) position.

Analysis of dorsal fin spines from waters off Taiwan

A preliminary comparison of four spines from 10 females and 15 males, collected from the waters off Taiwan from Sep–Oct 2016, found the clearest growth rings with minimal vascularization in the longest dorsal spine of the first dorsal fin. Based on these findings, and in reference to the recommendations of Shimose et al. (2015), the longest dorsal spine was selected as the primary sample for age determination, starting in Nov 2016, with the second-longest dorsal spine used as supplementary material. The dorsal fin spines extracted from 616 Pacific blue marlin were kept in cold storage until being boiled to remove tissue and to extract the longest spine (which tended to be the fifth or sixth spine) from the dorsal fin. Two 0.5-mm thick cross sections were taken successively along the length of each spine with a low-speed Isomet saw (Buehler 11–1280–160) with diamond wafering blades. Measuring up from the base of the spine, cuts were made approximately at a location equivalent to approximately half of the maximum width of the condyle base measured perpendicularly from the line of maximum condyle width (Sun et al. 2002, Chiang et al. 2004). Digital images of spine sections were captured under transmitted light at magnifications of $\times 10$ –25 using a DP22 Olympus camera attached to a SZ61 Olympus light microscope. Following the method described by Kopf et al. (2011), measure-

ments (± 0.01 mm) included spine radius (R_{MG}) and growth band radius (r_i), the maximum distances from the outer edge of spine section and the outer edge of each translucent zone i to the focus, respectively. These measurements are illustrated in Fig. 2. Measurements were recorded from digital images using free public-domain software, Image J version 1.51 K (<https://imagej.net/ij/>; Schneider et al. 2012).

Growth bands were visible as alternating wide opaque and narrow translucent zones, and age estimates were assigned based on the total number of paired opaque and translucent zones with reference to Kopf et al. (2011) and Shimose et al. (2015), referred to hereafter as “annual growth band.” Narrow translucent zones were delineated on digital images under transmitted light for age determination and annulus measurements, as they are considered to mark the presumed endpoint of annual growth increments (Fig. 3). In this study, we defined false growth bands as those were the translucent zone failed to extend into the cranial or caudal margin of the section, or if the translucent zone was faint, thin, or segmented (Speare 2003, DeMartini et al. 2007). False annual growth bands, characterized as seasonal rings formed prior to age 1, were identified based on the position of growth bands in spine sections for the individuals with otolith-based daily age estimates of <1 year (Fig. 3a, b, c, and f). For some specimens,

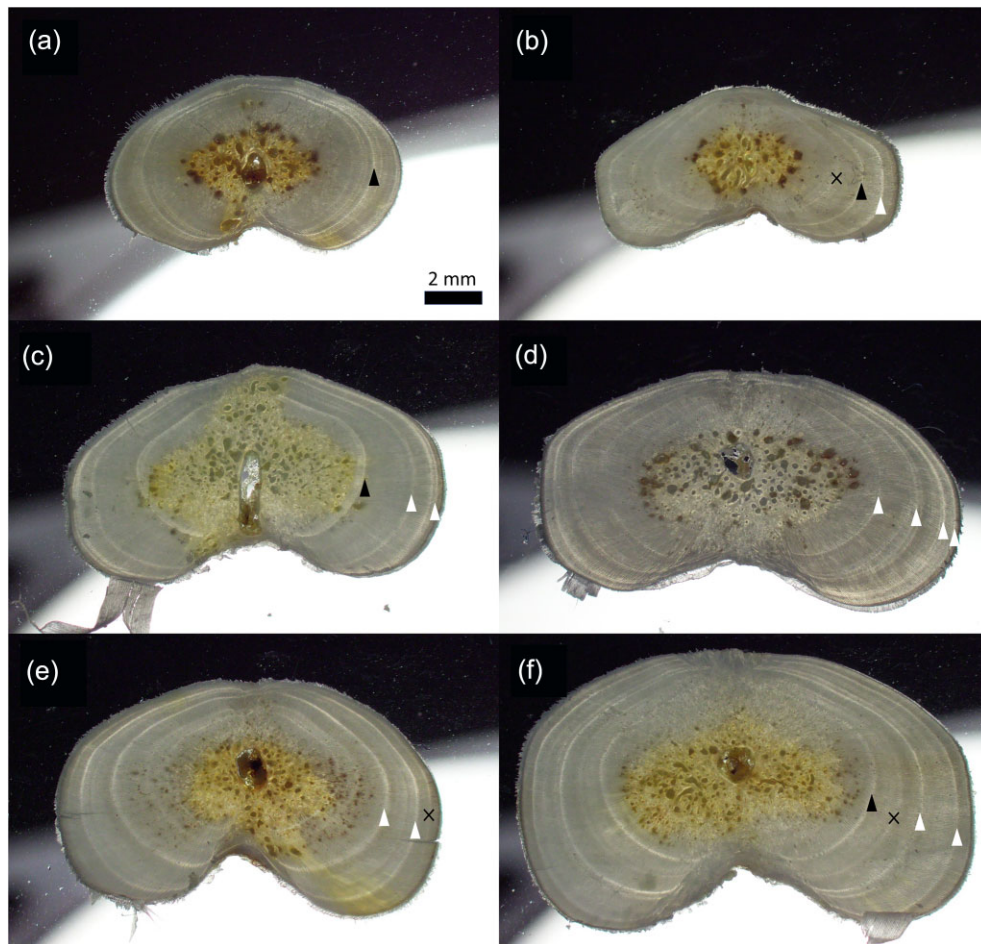


Figure 3. Transverse sections of dorsal spines from six individual Pacific blue marlin (*Makaira nigricans*), viewed under transmitted light: (a) 133 cm eye-to-fork length (EFL) male; (b) 152 cm male; (c) 169.8 cm male; (d) 175 cm male; (e) 181.6 cm female; and (f) 195.4 cm female. White triangles mark assumed annual growth bands, cross symbols indicate false growth bands, and black triangles denote bands presumed to have formed before 1 year of age.

the first several growth rings could be obscured by the vascularized core of the spine or not clearly visible in the absence of vascularization. The number of early but missing growth rings was therefore estimated using the replacement method of Hill et al. (1989) and Chiang et al. (2004). This involved first compiling ring radii statistics from younger specimens that had at least clearly visible first or second rings. Radii of the visible rings of the samples which had missing early rings were then compared with the data for younger specimens. When the smallest radius among the visible rings in a sample fell within one standard deviation of the mean radius of a ring from the younger specimens, the number of missing rings was estimated as the difference between the ring counts of the reference and the sample.

Although evidence from larval distributions suggests that blue marlin are capable of spawning year-round in the WC-NPO (Nishikawa et al. 1985), gonadal development and reproductive indicators suggest that the primary spawning season occurs between May and Sep (Shimose et al. 2009, Sun et al. 2009). Annual growth bands in sectioned spines are typically formed between Sep and Oct (Shimose et al. 2015, see the section “MIR analysis”). Consequently, the first growth band is assumed to form at ~ 0.42 years of age, corresponding to the period from hatching in mid-May to growth band formation in mid-Oct ($5/12 = 0.42$). This band is regarded

as a false annual growth band (Shimose et al. 2015). The second growth band, forming at ~ 1.42 years of age, is regarded as the first true annual growth band. All spine sections were read twice by the same experienced reader, with an interval of more than two weeks between the readings. The precision of the annual growth band counts was assessed using APE and CV, while modified age bias plots (Ogle et al. 2020) and *t*-tests (if the sample size was >3) were used to detect systematic disagreement between paired age readings.

Marginal increment ratio analysis

The periodicity and timing of translucent zone formation was determined using marginal increment ratio (MIR) analysis (Campana 2001). It was estimated for each specimen using the equation (Prince et al. 1988): $MIR = (R_{MG} - r_n)/(r_n - r_{n-1})$, where R_{MG} is the spine radius; and r_n the radius (mm) of the last annual growth band, and r_{n-1} the radius (mm) of the penultimate annual growth band. A one-way ANOVA on ranks (Kruskal-Wallis test, K-W test) (Zar 1999) was used to compare monthly differences in *MIR* across all age classes combined. The timing of annulus formation was compared statistically by grouped monthly *MIR* values using the Dunn's test to compare post hoc analyses (Zar 1999).

Table 2. Summary of specified prior distributions for the age determination-error von Bertalanffy growth model (VBGM) and two-stanza growth model (TSGM) in the Bayesian analysis.

Parameter	Description (unit)	Prior
VBGM		
L_∞	Asymptotic length (cm)	$L_\infty \sim U(30, 500)$
K	Brody growth rate coefficient (year ⁻¹)	$K \sim U(0.01, 4)$
t_0	Theoretical age at length is zero (year)	0
$t^\dagger$	True age (year)	$t \sim U(0, 50)$
$\sigma_L^{2\dagger}$	Observation error variance of length (cm)	$1/\sigma_L^2 \sim \text{Gamma}(2, 0.45)$
$\sigma_A^{2\dagger}$	Observation error variance of age (year)	$1/\sigma_A^2 \sim \text{Gamma}(2, 0.45)$
TSGM		
β_1	Gompertz growth parameter 1	$\beta_1 \sim U(50, 300)$
β_2	Gompertz growth parameter 2	$\beta_2 \sim U(0, 30)$
β_3	Gompertz growth parameter 3	$\beta_3 \sim U(1, 30)$
a	Schnute curve parameter	$a \sim U(1, 10)$
L_2	Length-at-age t_2 (cm)	$L_2 \sim U(1, 500)$

[†]Prior specifications were applied to both the VBGM and TSGM.

Decimal age algorithm

A decimal age for each individual was calculated based on the count of annual growth bands, along with the known primary spawning season (May–Sep; Shimose et al. 2009, Sun et al. 2009) and date of capture (Shimose et al. 2015). Fish captured in Sep and Oct were assumed to have completed at least one annual growth band since their last birthday, so no adjustment was made to the growth band count. Age readings of fish collected during Nov and Apr were adjusted by adding 0.5 years to the annual growth band count, while fish collected between May and Aug were aged by adding 1 year, as they had reached their birth months but the annual growth band had not yet formed.

Coupling the age determination error and the growth models

Sex-specific growth was modeled using VBGM and a TSGM based on a Bayesian errors-in-variables (EIV) approach (Hatch and Jiao 2016). Age-length data based on otoliths and fin spines included the samples from Taiwan waters and length-at-age records from the ISC IBBS Program (specimens from the WCNPO).

The age determination-error VBGM was described by the equation:

$$\begin{cases} L_i = L_\infty \times (1 - \exp(-K \times (t'_i - t_0))) e^{\varepsilon_i^L} \\ t'_i = t_i \times e^{\varepsilon_i^A} \end{cases} \quad (1)$$

where L_i is the observed length-at-age for the i th individual; L_∞ is the asymptotic length; K is the Brody growth coefficient; t_0 is the hypothetical length at age 0 (fixed at zero); t'_i is the observed age from the multiple age reads either by otolith micro-increment counts or by fin-spine annual growth band counts for the i th individual, t_i is the true age for the i th individual. The length observation error (ε_i^L) for the i th individual is assumed to be independent and normally distributed with mean 0 and variance σ_L^2 . For the age determination error for the i th individual (ε_i^A), the logarithm of observed age $\ln(t'_i)$ is assumed to be independent and normally distributed with mean $\ln(t_i)$ and variance σ_A^2 . The otolith daily-growth data was combined by sex during model fitting, following the approach of Shimose et al. (2015).

The age determination-error TSGM was described by the equations:

(1) Juvenile Gompertz growth function (for age = 0–1)

$$L_i = \beta_1 \times \exp(-\beta_2 \times \exp(-\beta_3 \times t'_i)) \times e^{\varepsilon_i^L} \quad (2)$$

where L_i is the observed length-at-age for the i th individual (females and males combined); β_1 , β_2 , and β_3 are model parameters. Based on exploratory analysis, age 1 was chosen as the transition point where growth begins to approach the asymptotic length under the Gompertz model.

(1) Subadult and adult Schnute growth function (Schnute 1981) (applied to ages 1–27 for females and ages 1–18 for males)

$$\begin{cases} L_i = L_1 \times \exp\left(\ln(L_2/L_1) \frac{1 - \exp(-a \times (t'_i - t_1))}{1 - \exp(-a \times (t_2 - t_1))}\right) \times e^{\varepsilon_i^L} \\ t'_i = t_i \times e^{\varepsilon_i^A} \end{cases} \quad (3)$$

where L_i is the observed length-at-age for the i th individual; L_1 is the predicted EFL-at-age one derived from the juvenile Gompertz growth function; and L_2 is the predicted EFL at age $t_2 = 27$ for females and age $t_2 = 18$ for males, corresponding to the maximum observed ages reported by Hill et al. (1989); a is model parameter which determines the shape of the growth curve; t'_i is the observed age for the i th individual. The length observation error (ε_i^L) and age determination error (ε_i^A) adhere to the same statistical distribution assumptions as the VBGM; however, the error variance will be re-estimated.

A Bayesian estimator was used to construct the joint posterior probability distribution for parameters in the growth models (Hatch and Jiao 2016). A list of the non-informative prior distributions used for the other parameters for each model is provided in Table 2. Model parameters were estimated using MCMC simulation with the Gibbs sampling algorithm as implemented in the WinBUGS 1.4.3 software (Spiegelhalter et al. 2007). For the TSGM, the posterior distribution of parameters of the Gompertz and Schnute models were jointly estimated in the MCMC simulations. MCMC simulations were conducted by simulating three chains. For each chain, 333 330 iterations sampled with a thinning rate of 1/25 and a burn-in period of 83 332 iterations for a total of 10 000 MCMC samples. Convergence of the MCMC samples to the posterior distribution was checked for each model by monitoring the trace and autocorrelation of model parameters. The Gelman and Rubin (1992) and Heidelberger and Welch (1983) diagnostics as implemented in the R language

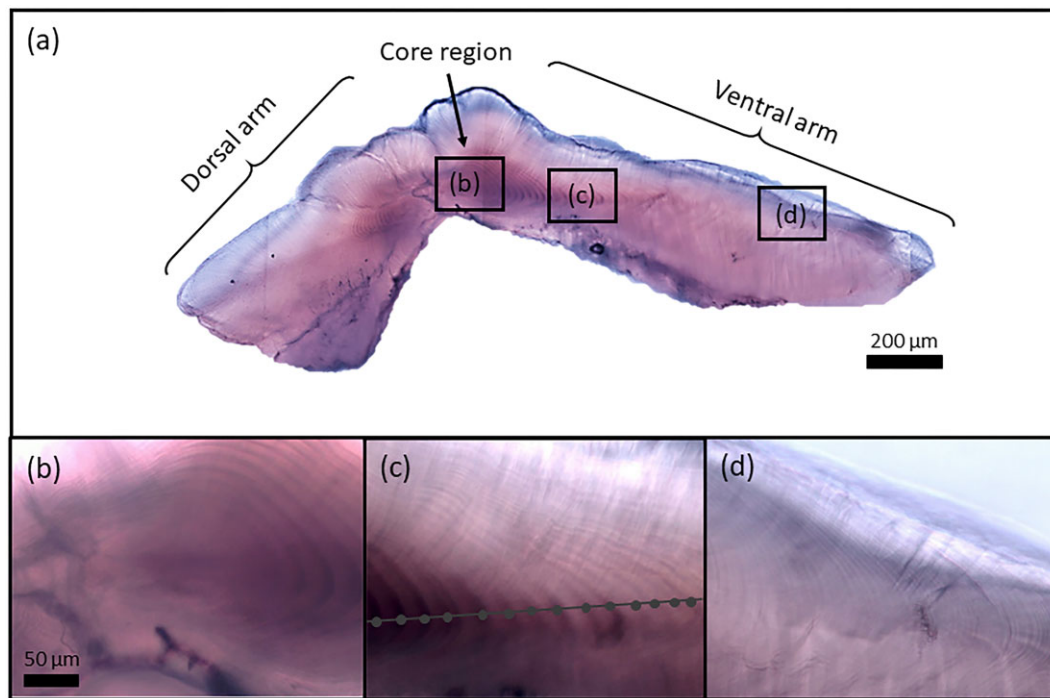


Figure 4. Photographs of (a) a transversely sectioned sagittal otolith viewed under transmitted light for daily age reading and (b–d) three magnified areas from a 138 cm eye-to-fork length (EFL) male Pacific blue marlin (*Makaira nigricans*). Presumed daily micro-increments are marked by gray circles in panel (c).

(R Development Core Team 2018) and the CODA package (Best et al. 1995) were also examined. The deviance information criterion (DIC, Spiegelhalter et al. 2002) was used to select the most parsimonious model among the Bayesian models. In this study, the model with the lowest DIC was considered the best model between the VBG and TSGM applied to the Pacific blue marlin.

Results

Otolith micro-increments

In the otolith analysis of juvenile blue marlins from Taiwan waters, micro-increments were observed from the core to the edge of the ventral arm, but no micro-increments were visible in the dorsal arm for age determination. Around the core region, visual enhancement of micro-increments required fine focal adjustment of the microscope (Fig. 4b). Micro-increments were reasonably distinct between the core region and the ventral arm before inflection (Fig. 4c) and were enumerated using magnifications of 200× and 400×. Daily age estimates were confidently determined for 13 (115–152 cm EFL) of 25 juvenile individuals, with age counts ranging from 213 to 349 days. Reliable enumeration was limited to this range, as micro-increments became minute and discontinuous beyond the ventral arm inflection in the otolith section (at ~1 year of age). The mean APE was 4.44%, and the CV was 6.27%. The back-calculated hatch dates for eight specimens (62%) occurred within the primary spawning season (May–Sep), while the remaining specimens were distributed across the remainder of the year.

For the combined otolith age estimation data ($n = 40$), encompassing samples from Taiwan waters and data from the

ISC IBBS Program [i.e. additional 27 otolith samples from Shimose et al. (2015)], daily age estimates ranged from 26 to 349 days, and EFL measurements ranged from 17.2 to 159.7 cm (Fig. 5). Length-at-age was fitted to a Gompertz growth function [i.e. Eq. (3)] of $L_t = 147.81 \times \exp(-2.96 \times \exp(-6.20 \times t))$. No significant differences in length-at-age between sexes were observed before 1 year of age. The model predicted EFLs of 129.36 cm and 146.92 cm at ages 0.5 and 1 year, respectively.

Annual growth bands in fin spines

From Taiwan samples, 504 fin spine sections were successfully interpreted, with readability rates of 83% for females and 80% for males. The reading precision, measured as average APE and CV, was 10.77% and 15.23% for females, and 9.23% and 13.06% for males, respectively. False annual growth bands were identified based on the position of growth bands in spine sections for 8 of 13 male individuals with otolith-based daily age estimates of <1 year and were observed over a radius range of 3.71 to 4.26 mm, with a modal frequency predicted at 4.02 ± 0.15 mm. A total of 113 female spines and 132 male spines had at least the first annual growth band visible. The mean radius of each annual growth band for females and males appears in Tables 3 and 4, respectively. The mean radius of the annual growth band increased linearly with age for both sexes. For females, the mean radius increased from 5.23 ± 0.40 mm at the first annual growth band to 12.78 ± 0.16 mm at the 10th, while for males, it increased from 5.10 ± 0.26 mm to 10.33 mm over the same interval (Fig. 6).

The results of multiple readings, shown in Supplementary Fig. S1, indicate that most of the observed values fall along

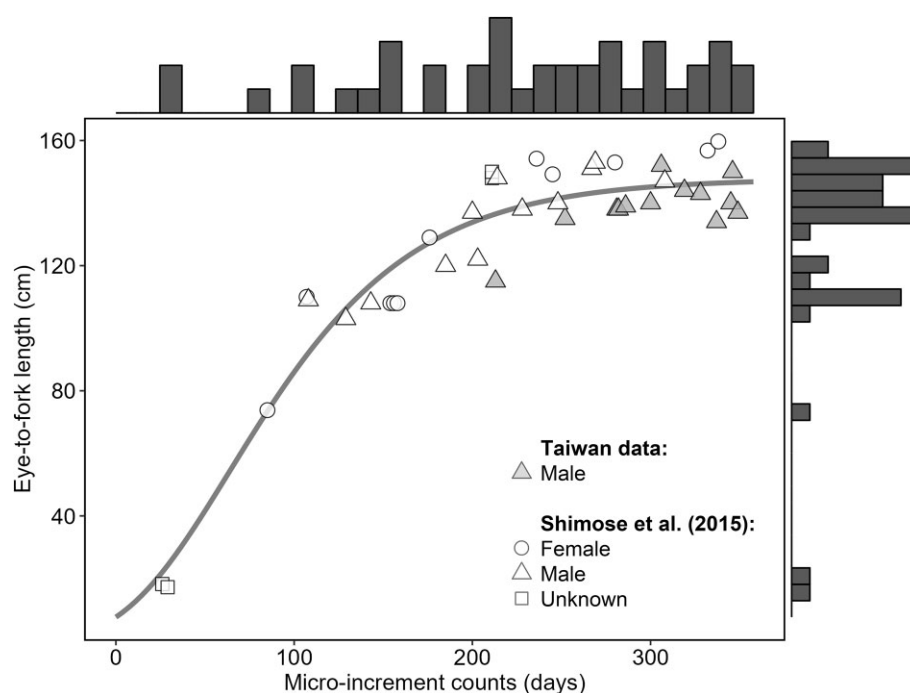


Figure 5. Scatter plot of eye-to-fork length (cm) versus micro-increment counts (days), overlaid with the estimated Gompertz growth curve for Pacific blue marlin (*Makaira nigricans*). Marginal histograms on the top and right display the distributions of daily age and length, respectively. Data points represent unknown sex individuals (squares), females (circles), and males (triangles), based on samples collected in the waters off Taiwan and those reported by Shimose et al. (2015) in the western and central North Pacific Ocean (compiled by the ISC IBBS Program; see Table 1).

Table 3. Mean radius (mm) and standard deviation of annual growth bands for female Pacific blue marlin (*Makaira nigricans*) from samples collected in the waters off Taiwan. Roman numerals denote the number of annual growth bands. Numbers in parentheses are the number of specimens for which the specified ring was readable. “—” indicates that data is not available.

<i>n</i>	Age class	I	II	III	IV	V	VI	VII	VIII	IX	X
17	1	5.67, 0.60 (17)									
96	2	5.20, 0.28 (53)	6.69, 0.46 (96)								
76	3	5.19, 0.30 (25)	6.73, 0.34 (69)	8.04, 0.31 (76)							
36	4	5.02, 0.25 (10)	6.75, 0.28 (29)	8.29, 0.34 (30)	8.76, 0.34 (36)						
29	5	4.85, 0.26 (7)	6.71, 0.30 (16)	8.27, 0.28 (19)	8.82, 0.39 (21)	9.38, 0.35 (29)					
8	6	5.29,—(1)	6.80, 0.25 (3)	8.10, 0.64 (4)	8.96, 0.50 (6)	9.52, 0.30 (6)	10.14, 0.13 (8)				
3	7	—	6.36,—(1)	7.65, 0.67 (2)	8.66, 0.29 (2)	9.27,—(1)	9.92, 0.48 (3)	10.60, 0.32 (3)			
9	8	—	6.72,—(1)	8.35, 0.10 (5)	8.78, 0.33 (3)	9.48, 0.18 (5)	10.24, 0.14 (3)	11.06, 0.31 (5)	11.34, 0.45 (9)		
1	9	—	—	—	9.08,—(1)	—	9.85,—(1)	—	11.13,—(1)	11.94,— (1)	
2	10	—	—	8.12,—(1)			10.18,—(1)	10.93,—(1)	11.3, 0.76 (2)	—	12.78, 0.16 (2)
277	Sum	5.23, 0.40 (113)	6.71, 0.39 (215)	8.13, 0.34 (137)	8.8, 0.36 (69)	9.41, 0.32 (41)	10.10, 0.24 (16)	10.89, 0.35 (9)	11.32, 0.45 (12)	11.94,— (1)	12.78, 0.16 (2)

the diagonal line, suggesting that the data from the two readings are very similar. There was no statistically significant bias in the age estimates between the multiple readings. Thereafter, the first reading was chosen for subsequent data analysis.

MIR analysis

The MIRs of 290 specimens from Taiwan samples were usable, but the remaining spine sections had fewer than two consecutive annual growth bands or low readability. The mean MIR showed a gently fluctuating trend over the course of 1

Table 4. Mean radius (mm) and standard deviation of annual growth bands for male Pacific blue marlin (*Makaira nigricans*) from samples collected in the waters off Taiwan. Roman numerals denote the number of annual growth bands. Numbers in parentheses are the number of specimens for which the specified ring was readable. “-” indicates that data is not available. Six of the 214 individuals were not included in this table, as it excludes specimens whose spine sections showed only one band, which was identified as a false annual growth band.

<i>n</i>	Age class	I	II	III	IV	V	VI	VII	VIII	IX	X
38	1	5.00, 0.23 (38)									
72	2	5.10, 0.22 (51)	5.73, 0.20 (72)								
50	3	5.13, 0.31 (26)	5.82, 0.28 (29)	6.29, 0.26 (50)							
29	4	5.22, 0.34 (12)	5.87, 0.23 (21)	6.47, 0.26 (23)	7.00, 0.24 (29)						
8	5	5.35, 0.18 (3)	-	6.85, 0.21 (3)	7.20, 0.15 (4)	7.72, 0.10 (8)					
6	6	5.17, 0.42 (2)	6.28,—(1)	6.16,—(1)	7.43, 0.03 (2)	7.50, 0.30 (4)	7.96, 0.23 (6)				
3	7	-	-	6.61,—(1)	7.37, 0.09 (2)	7.66, 0.36 (3)	8.10, 0.09 (2)	8.43, 0.16 (3)			
1	8	-	-	-	-	-	-	8.34,—(1)	8.79,— (1)		
-	9	-	-	-	-	-	-	-	-	-	-
1	10	-	-	-	-	-	-	-	-	9.83,— (1)	10.33,— (1)
208	Sum	5.10, 0.26 (132)	5.78, 0.23 (123)	6.37, 0.29 (78)	7.06, 0.26 (37)	7.65, 0.23 (15)	7.99, 0.20 (8)	8.41, 0.14 (4)	8.79,— (1)	9.83,— (1)	10.33,— (1)

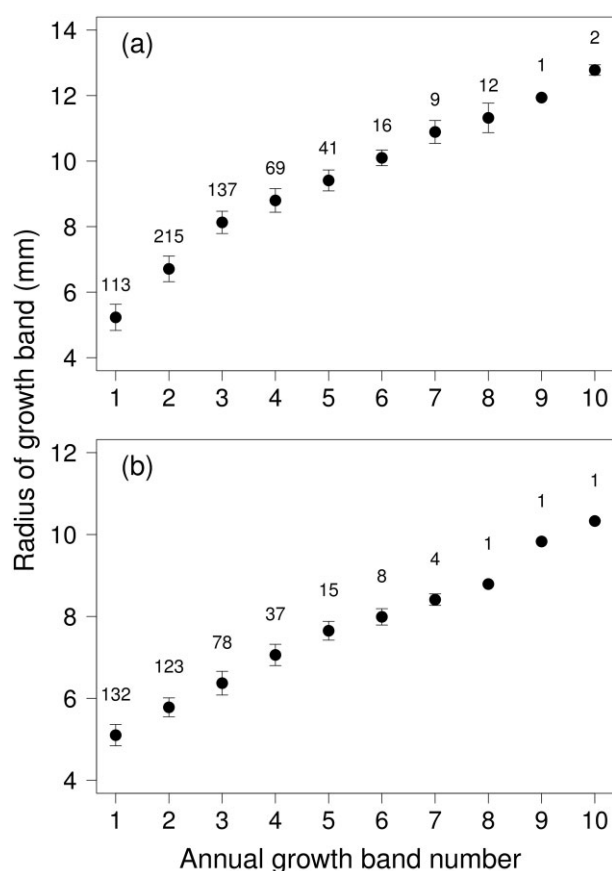


Figure 6. Mean ($\pm$ standard deviation, SD) radius of growth bands for (a) female and (b) male blue marlin (*Makaira nigricans*) from samples collected in the waters off Taiwan. The numbers above the vertical bars represent the sample sizes.

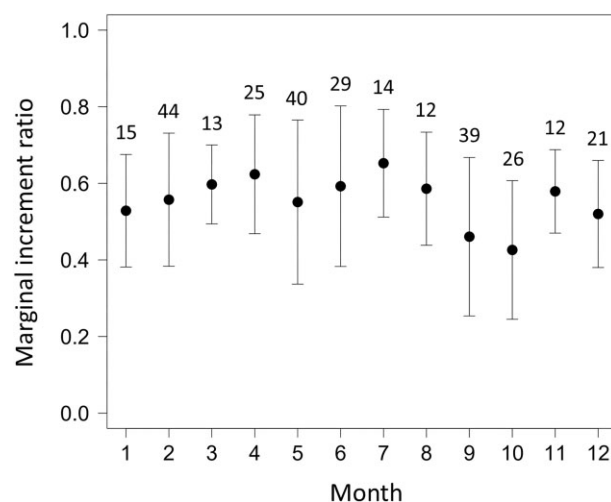


Figure 7. Monthly changes in the mean marginal increment ratio of blue marlin (*Makaira nigricans*) (combined sexes and years) from samples collected in the waters off Taiwan. Vertical bars represent the standard deviation (SD), and the numbers on the bars indicate the sample sizes.

year for the combined age samples (Fig. 7). However, due to the small sample size, it was not possible to separate MIR values by age group or gender. Specifically, the monthly variation in MIR was higher from Mar to Apr (0.60–0.62) and Jun to Aug (0.59–0.65), with the lowest values observed in Sep and Oct (0.43–0.46), followed by an increase in Nov (0.58). It is inferred that a new annulus is formed between Sep and Oct each year on the dorsal fin spine of Pacific blue marlin. There is a significant difference in MIR values among months (K–W test; $P < 0.001$). The median MIR in Sep and Oct was, respectively, significantly lower than in the other months of the year (Dunn’s test; $P < 0.05$).

Table 5. Posterior median of the sex-specific growth parameters derived from the age determination-error von Bertalanffy growth model (VBGM) and two-stanza growth model (TSGM) for the Pacific blue marlin (*Makaira nigricans*). Numbers in parentheses are the standard deviations (SD). “-” means value not available. DIC denotes Deviance Information Criterion.

Parameter	VBGM		TSGM	
	Female	Male	Female	Male
L_{∞}	204.8 (1.35)	163.6 (0.65)		
K	1.56 (0.06)	2.39 (0.07)		
t_0	0 (-)	0 (-)		
β_1			151.6 (2.48)	148.9 (1.02)
β_2			3.45 (0.150)	3.49 (0.134)
β_3			6.44 (0.390)	6.70 (0.312)
t_1			1 (-)	1 (-)
t_2			27 (-)	18 (-)
$L_1^{\dagger}$	161.2 (2.16)	148.5 (0.86)	150.8 (2.27)	148.2 (0.94)
L_2			267.2 (7.45)	198.6 (6.07)
a			0.24 (0.026)	0.23 (0.045)
σ_A	0.13 (0.006)	0.19 (0.010)	0.13 (0.005)	0.19 (0.009)
σ_L	0.12 (0.004)	0.09 (0.002)	0.08 (0.003)	0.07 (0.002)
DIC	5016	6628	4565	6335

[†]Values represent the estimated derived quantities of length-at-age one from the models.

Age estimation of fin spines

Female and male blue marlin showed similar observed age ranges for the Taiwanese port-sampled fish. Among the Taiwan samples (excluding 13 samples younger than 1 year by otolith daily micro-increment age estimation), the estimated ages of 277 female specimens ranged from 1.5 to 11 years, and EFL measurements ranged from 142.8 to 269.4 cm. For 214 males, estimated ages ranged from 1 to 10.5 years, and EFL measurements between 133 and 208 cm (Supplementary Tables S2 and S3).

For the combined spine age estimation data, which includes specimens from the waters off Taiwan and the ISC IBBS Program (covering the WCNPO; see Fig. 1 and Table 1), age estimates were higher for female blue marlin. For females, ages ranged from 1 to 27 years, and EFL measurements ranged from 136 to 330.5 cm. For males, estimated ages ranged from 1 to 18 years, and EFL measurements between 118 and 232.1 cm. The combined data showed that blue marlin growth was sexually dimorphic. Female growth patterns were characterized by rapid initial growth, reaching ~170 cm by age 2 and surpassing 240 cm by age 9, after which growth slowed. Similarly, males exhibited rapid early growth, reaching 158 cm by age 2 and 180 cm by age 6.5, with growth rates declining thereafter (Supplementary Tables S2 and S3). Additionally, the mean length-at-age after 1 year of age was consistently higher in females than in males.

Growth model

The Gelman and Rubin statistics for all parameters, including all the variance terms, equaled 1.0, which provided no evidence for lack of convergence of the Markov chain for both VBGM and TSGM. The parameters also passed the Heidelberg and Welch stationarity and halfwidth tests. The trace plots of the major parameters did not indicate lack of convergence for both models (Supplementary Figs S2 and S3).

The posterior distributions of the parameters of the VBGM and TSGM for both sexes are presented in Table 5 and Supplementary Figs S4 and S5. For the VBGM, the estimated posterior medians of L_{∞} and K were 204.8 cm EFL (95%

credible intervals [CrI]: 202.1–207.4 cm) and 1.56 year⁻¹ (95% CrI: 1.45–1.68 year⁻¹) for females, and 163.6 cm EFL (95% CrI: 162.3–164.9 cm) and 2.39 year⁻¹ (95% CrI: 2.26–2.53 year⁻¹) for males. For the TSGM, the posterior medians of L_1 were 150.8 cm EFL (95% CrI: 146.4–155.2 cm) for females and 148.2 cm EFL (95% CrI: 146.3–150.0 cm) for males, with slightly greater uncertainty observed in the estimates for females. Additionally, the EFL-at-age one (146.92 cm) estimated from otolith micro-increments closely matched the TSGM prediction (150.8 cm for female, 148.2 cm for male) but differed from the VBGM-predicted value for males (EFL-at-age one: 161.2 cm for females and 148.5 cm for males). L_2 was 267.2 cm EFL (95% CrI: 253.9–282.9 cm) for females and 198.6 cm EFL (95% CrI: 188.3–212.0 cm) for males. The shape parameter (a) was slightly higher for females (0.24) than males (0.23), with overlap in their distributions. Age determination error standard deviation (σ_A) was slightly higher for males (0.19) than for females (0.13) in both VBGM and TSGM.

The results indicated that the growth curve (posterior median) of the VBGM, along with its 95% CrI, consistently underestimated the lengths of older individuals, particularly those >5 years of age in both sexes (Fig. 8a and c). In contrast, the growth curve of TSGM and its 95% CrI provided a better fit to the observed data at older ages, effectively capturing the continued increase in body length with increasing age (Fig. 8b and d). Moreover, the 95% CrI of VBGM were narrower than those derived from the TSGM. The 95% prediction intervals (PI) derived from the VBGM systematically underestimated growth for both sexes, with several observed values falling outside the predicted bounds (e.g. males exceeding 195 cm EFL; females exceeding 260 cm EFL; Fig. 8a and c). In contrast, the 95% PI estimated by the TSGM more accurately captured individual variability in growth (Fig. 8b and d). Notably, one male (232.1 cm EFL at age 14) and two female (330.5 and 321.9 cm EFL at age 20 and 27) individuals at greater ages fell outside the 95% PI. Furthermore, the smaller DIC values for the TSGM in both sexes (DIC = 4 565 for females and 6 335 for males) further supported its selection as the most parsimonious model, outperforming the VBGM (DIC = 5 016 for females and 6 628 for males).

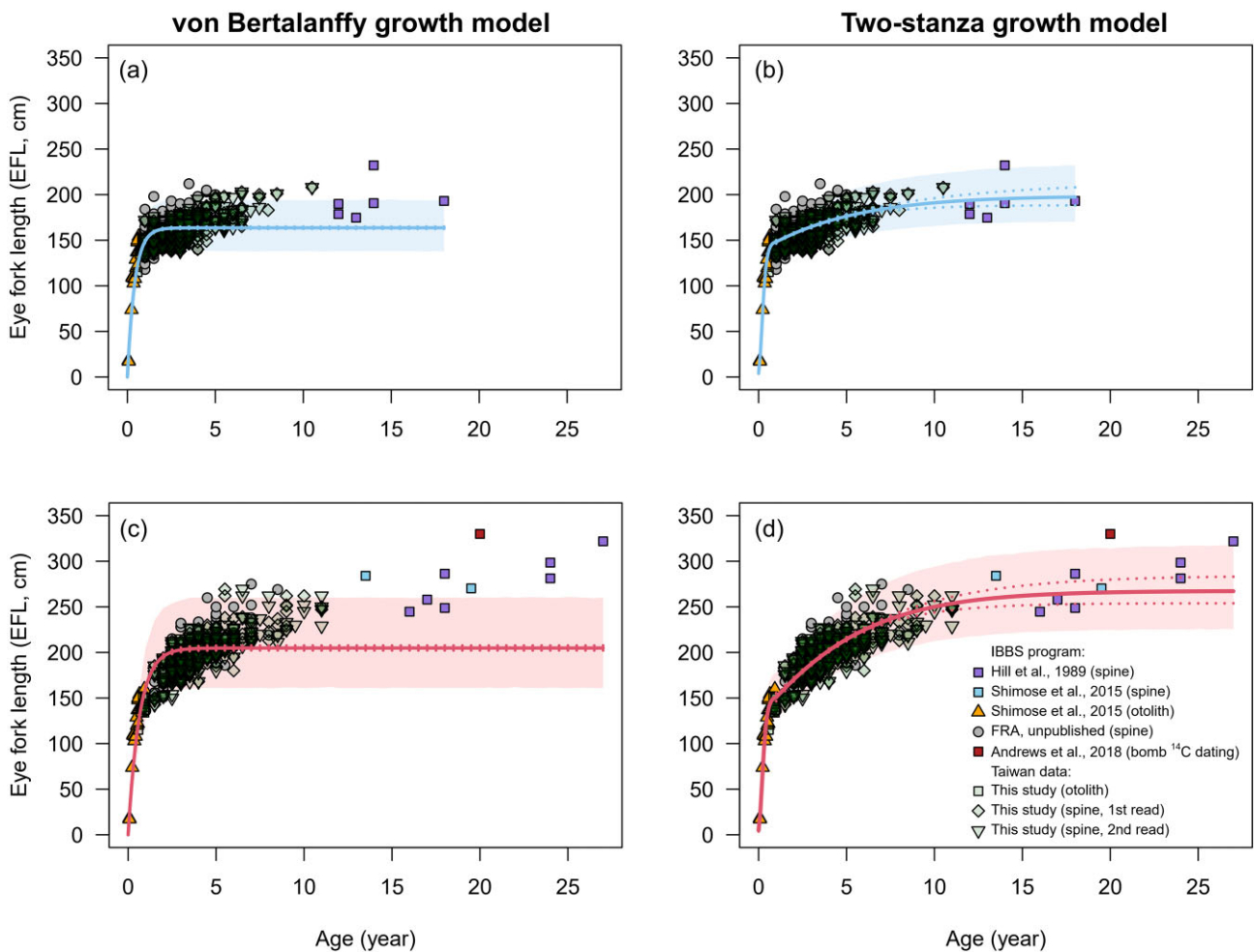


Figure 8. Growth curves (posterior medians; solid lines) estimated using the age determination-error von Bertalanffy growth model (left panels) and the two-stanza growth model (right panels) for male (a and b) and female (c and d) Pacific blue marlin (*Makaira nigricans*), based on length-at-age data combining samples collected in the waters off Taiwan with data from the ISC IBBS Program (specimens from the western and central North Pacific Ocean). The 95% credible intervals of the growth curve estimates are shown as dotted lines, and the 95% prediction intervals are depicted as shaded polygons.

Discussion

Otolith micro-increment counts have been previously used to validate the position of the first annual growth mark in fin spines of billfishes (e.g. DeMartini et al. 2007, Kopf et al. 2011, Shimose et al. 2015). Prince et al. (1991) and Kopf et al. (2011) compared the back-calculated hatch dates of Atlantic blue marlin (*Makaira nigricans*) and South Pacific striped marlin (*Kajikia audax*), respectively, with their known spawning seasons, finding that the assumption of one micro-increment forming per day was most consistent with the results. Similarly, Shimose et al. (2015) demonstrated that the back-calculated hatch dates of Pacific blue marlin generally aligned with the known primary spawning season from May to Sep (Shimose et al. 2009, Sun et al. 2009), accounting for 59% of all samples, supporting the validity of using micro-increment counts to estimate daily age. Consistent with these findings, our study revealed that 8 of the 13 otolith specimens under 1 year old (62%; 115–152 cm EFL, 213–319 days) had back-calculated hatch dates within the primary spawning season (May to Sep). The remaining five specimens, measuring 134 to 150 cm EFL and aged 328 to 349 days, had hatch dates distributed outside this period. However, it should be

noted that age underestimation may occur when using otolith micro-increment counts in billfishes, particularly in juveniles approaching or exceeding 1 year of age. In such cases, sub-micron increments, especially in the outer otolith region, can be difficult to resolve under light microscopy, particularly if section preparation is suboptimal. Speare (2003) examined otolith micro-increments from a tetracycline-injected black marlin (*Istiompax indica*, 150 cm LJFL), enumerated a total of 452 increments, and found that 178 were located outside the tetracycline mark, closely matching the known 198 days at liberty with a slight (~5%) underestimation. This discrepancy may arise from several factors, including variability in increment formation, challenges in enumerating faint or compressed micro-increments, and limitations in otolith preparation and reading techniques. Future studies employing sagittal otolith pairs prepared separately for light microscopy and scanning electron microscopy (SEM) may enhance increment count accuracy and help identify potential underestimation in older juveniles of Pacific blue marlin. These challenges also underscore the importance of validation experiments, such as oxytetracycline-based age validation (Speare 2003), to enhance the reliability of age estimates derived from otolith micro-increments.

Given the constraints of otolith daily micro-increment age estimation beyond the first year of Pacific blue marlin, Shimose *et al.* (2015) advocated for the use of fin spines when assessing the age of older individuals, an approach that was adopted in the present study. Notably, our study detected false annual growth bands in the spine sections of eight of 13 male specimens that were estimated to be <1 year old based on otolith micro-increment counts. These early-formed bands in the fin spines may lead to age overestimation if misinterpreted as true annuli. Therefore, integrating otolith micro-increment analysis with fin spine-based age estimation is critical to minimizing age determination errors in Pacific blue marlin and other billfishes (DeMartini *et al.* 2007, Kopf *et al.* 2011, Shimose *et al.* 2015).

MIR analysis indicates that growth rings in Pacific blue marlin are formed once a year, typically between Sep and Oct, after the peak spawning season from May to Sep. This timing aligns with findings from Shimose *et al.* (2015) and is biologically reasonable, as post-spawning energy depletion and seasonal environmental changes (e.g. variations in seawater temperature) likely lead to slowed growth and annulus formation. While many MIR results on billfish have concluded that growth bands in sectioned fin spines are formed annually (Kopf *et al.* 2011, Sun *et al.* 2015), some have identified two decreasing trends in MIR, suggesting the potential for two cycles per year (Chiang *et al.* 2004, DeMartini *et al.* 2007). One of these may represent a seasonal false annulus, underscoring the need for a more detailed understanding of annulus formation dynamics in Pacific blue marlin.

Farley *et al.* (2022) conducted a direct comparison of age estimates derived from annual growth bands in fin spines and otoliths of swordfish in the southwest Pacific Ocean, revealing a clear bias in which spine ages were consistently lower than otolith ages for females older than 7 years and males older than 4 years. This divergence in age counts between annuli in otoliths and fin spines is likely attributable to the remodeling and resorption of fin spines, particularly in older individuals. Although fin spines are widely utilized for age estimation of billfish due to their ease of collection and preparation, their reliability is challenged by factors such as the occurrence of false annual growth bands, vascularization, resorption, and limited reading precision. These issues indicate that the age estimates for Pacific blue marlin derived from fin spine sections in our study may be biased. While our previous exploratory analysis (based on the method of Farley *et al.* 2022) did not reveal clear annual growth bands in the otoliths of Pacific blue marlin, we recommend that future research overcomes the technical challenges of otolith annual age estimation and conducts a direct comparison of annual age estimates obtained from fin spines and otoliths. Such efforts are critical for determining the most reliable hard structure to provide accurate annual age estimates for this species.

Core vascularization in the fin spines of billfishes and swordfish, particularly in larger individuals, poses a significant challenge to accurate age estimation, as it frequently obscures the early annual growth bands (Drew *et al.* 2006, Chiang *et al.* 2004, Kopf *et al.* 2010, Garcia *et al.* 2017). Although several correction methods have been proposed to reconstruct lost annuli (e.g. Hill *et al.* 1989, Hoolihan *et al.* 2019), their applicability may warrant further refinement. The replacement method described by Hill *et al.* (1989) and Chiang *et al.* (2004), which estimates missing early rings by matching ring radii to those of younger specimens within 95% confidence

limits, may be limited by its reliance on the quality and representativeness of the reference dataset and its susceptibility to measurement error. Hoolihan *et al.* (2019) expanded on Hill *et al.* (1989) by assuming a power-function relationship between annulus radius and annulus number across individuals. However, substantial overlap in ring radii across annulus numbers limits the ability to statistically assign a radius to a specific annulus. Moreover, false annuli, such as doublets, triplets, and subannual increments, are frequently reported in both istiophorid billfishes and swordfish (Hill *et al.* 1989, Cayre and Diouf 1983, Kopf *et al.* 2010, 2011, Pons *et al.* 2016, DeMartini *et al.* 2007). If misclassified as true annuli and included in the correction process, these features will inflate age estimates and reduce the accuracy of correction methods such as that of Hoolihan *et al.* (2019). In light of these concerns, and considering that dorsal fin spine vascularization in Pacific blue marlin is generally minor and limited in extent (Shimose *et al.* 2015), this study adopted a simpler approach based on Hill *et al.* (1989) and Chiang *et al.* (2004) that avoids using an empirical formula to back-calculate the number of annuli requiring correction. A detailed comparison of these two correction methods is beyond the scope of the present study. We recommend that future research focus on developing more robust statistical approaches to increase confidence in age estimates derived from vascularized spines.

The present study describes the growth of Pacific blue marlin from juvenile to adult stages, incorporating both Taiwan samples and data from the ISC IBBS Program (specimens from the WCNP). For the combined otolith age estimation data ($n = 40$), the estimated EFL-at-age 1 year is 146.92 cm, which is smaller than the 161.8 cm EFL estimated by Shimose *et al.* (2015). This discrepancy may be due to the lower daily age estimates of otoliths for Taiwan samples in this study compared to those of Shimose *et al.* (2015). The observed variability in estimates may be attributed to the limitations of daily micro-increment age estimation, including reader subjectivity, difficulty in enumerating faint or compressed increments, and environmental factors affecting otolith formation. Despite this, the present study incorporates 40 otolith samples, addressing the limited juvenile sample size (beyond 0.7 years) in Shimose *et al.* (2015), and thereby provides a representative estimate of the average EFL for 1-year-old Pacific blue marlin.

This study utilized a TSGM to analyze the growth patterns of Pacific blue marlin, revealing significant changes in growth rates between sexes after 1 year of age. While Shimose *et al.* (2015) did not use a TSGM, their findings similarly indicated such changes. Based on the estimated minimum size at sexual maturity (157.8 cm EFL for females and 131 cm EFL for males; Sun *et al.* 2009), the observed growth shifts likely correspond to sexual maturation. Unlike traditional growth models, which assume a monotonic decline in growth rate with age, recent research on tunas and billfishes has highlighted notable growth transitions during life stages (Hearn and Polacheck 2003, Eveson *et al.* 2015). These transitions suggest at least two distinct growth phases: an initial rapid growth rate that slows during a transitional phase before stabilizing and gradually declining toward maximum length. To address the challenges of modeling such growth patterns, this study combined the Gompertz model, effective for describing rapid juvenile growth, with the Schnute model, which captures the gradual decline in older fish growth rates. The resulting two-stanza model showed superior statistical fit (lower DIC values) compared to the traditional VBGM and avoided underesti-

maturing asymptotic length. This combined approach offers a robust framework for describing Pacific blue marlin growth, and it is recommended for future research on other billfish species.

Shimose et al. (2015) estimated the maximum age of female Pacific blue marlin to be 19.5 years (with an EFL of 270.3 cm), with only three individuals exceeding 10 years of age. Andrews et al. (2018) validated the age of a large female specimen (EFL of 330.5 cm) from Hawaiian waters using bomb radiocarbon dating, estimating the most likely age to be 20 years. Hill et al. (1989) determined the maximum age of the largest-ever-studied female Pacific blue marlin to be 22 years at 398.8 cm EFL, with the oldest individual reaching 27 years at ~321.9 cm EFL. However, potential overestimation of age in the Hill et al. (1989) study—likely due to the overcounting of early growth bands during the first year—has been noted and addressed through otolith daily increment validation (Prince et al. 1991, Shimose et al. 2015), raising the possibility that the 27-year estimate could be inflated by a few years. Despite methodological differences, the maximum age estimates for female Pacific blue marlin across studies (Hill et al. 1989, Wilson et al. 1991, Shimose et al. 2015) are generally consistent, suggesting that the average maximum age of females is likely around 20 years. This study estimated the maximum age of male blue marlin from Taiwan waters to be 10.5 years (208 cm EFL), which is slightly higher than the 7.5 years at 185.8 cm EFL reported by Shimose et al. (2015). However, this 10.5-year estimate may still be underestimated, as predicted longevity is thought to be similar between sexes (Wilson et al. 1991), and Hill et al. (1989) reported a maximum estimated age of 18 years for males. Given the substantial discrepancies in maximum age estimates across studies, future research is recommended to validate the longevity of males using bomb radiocarbon dating, as demonstrated by Andrews et al. (2018).

Shimose et al. (2015) estimated L_{∞} at 264.3 cm and 186.3 cm EFL for females and males, respectively. In this study, the TSGM estimated maximum average lengths (L_2) of 267.2 cm EFL for females and 198.6 cm EFL for males, aligning with Shimose et al.'s findings. In contrast, Hill et al. (1989) reported a maximum observed length of 398.8 cm EFL for a female specimen, which differs substantially from the L_2 estimates obtained in this study. However, due to inconsistencies in the reported length and mass of that specimen, it was excluded from the present growth analysis. Notably, the extreme size and disproportional length-weight relationship of that specimen, along with one male (232.1 cm EFL at age 14) and two female individuals (330.5 and 321.9 cm EFL at ages 20 and 27, respectively) that fell outside the 95% PI of the TSGM in the present study, may also reflect the considerable growth plasticity of Pacific blue marlin (Andrews et al. 2018). In addition, size composition data from Japanese tuna longline fisheries, where individual ages are not known, indicate that most Pacific blue marlin are caught at lengths of ~200 cm EFL for females and 160 cm EFL for males. Specimens exceeding 300 cm (females) or 250 cm EFL (males) are exceedingly rare (Ijima and Shiozaki 2016). Relative to these fishery observations, the L_2 estimates in this study represent ~89% of 300 cm EFL for females and 80% of 250 cm EFL for males. Furthermore, the female L_2 corresponds to about 81% of the largest aged individual included in the growth curve analysis (330.5 cm EFL at 20 years; Andrews et al. 2018), while the male L_2 corresponds to about 86% of the largest

aged male (232.1 cm EFL at 14 years; Hill et al. 1989). It may also be worthwhile to consider the potential effects of more than six decades of fishing pressure on Pacific blue marlin (ISC 2021), which has likely driven size-selective exploitation and contributed to the decline in larger, older individuals, in recent catches (ISC 2021). This selection bias may account for the underrepresentation of older, more variable individuals in contemporary length samples, and has important implications (e.g. fishing-induced evolution of growth; Hilborn and Minte-Vera 2008, Enberg et al. 2012) for understanding natural growth trajectories.

Age determination errors pose significant challenges in fish stock assessment, as accurate age-length relationships are essential for deriving biological reference points from age-structured population models. In this study, multiple readings of spine growth bands were used to estimate the variance associated with age determination error (σ_A^2). Hatch and Jiao (2016) demonstrated that multiple age reads improve the estimation of age uncertainty, and simulations showed that not accounting for age determination errors results in more biased variability in predicted lengths. Given the challenges in age estimation of Pacific blue marlin, incorporating age determination errors into growth models helps address uncertainties in age estimation. This study recommends using measurement-error growth models for estimating blue marlin growth.

This study provides one of the most comprehensive analyses of the age and growth of Pacific blue marlin to date. It combines fin spine samples collected from Taiwan waters with historical length-at-age data from the IBBS program (specimens from the WCNP), covering a broader size range and spatial distribution than previous studies on this species. The TSGM offers a better fit than the traditional von Bertalanffy model, presenting a clearer depiction of growth patterns across life stages. This study also recommends the use of measurement-error growth models to account for variability in age determinations and enhance the accuracy of parameter estimation. While the predicted maximum average lengths align with those reported by Shimose et al. (2015), further comparisons with otolith-based annual ring age estimation are necessary to refine age estimates. It should be noted that the current sampling was geographically constrained to the WCNP. Future research should expand sampling to other Pacific regions, investigate potential spatial variation in growth, and incorporate additional validation techniques such as bomb radiocarbon dating and mark-recapture of tetracycline-injected individuals, to further improve our understanding of the age and growth of Pacific blue marlin.

Acknowledgements

We thank the observers for collecting blue marlin samples from fishing ports in Taiwan. We also express our gratitude to the Japanese fishers, market personnel, and crew members of the research and training vessels for their contributions to sample collection. We sincerely appreciate the valuable comments provided by two anonymous reviewers, as well as by Jiangang Luo, John Hoolihan, and Editor Allen Andrews. We also thank Zi-Wei Yeh for his work in improving the figures.

Author contributions

Y.J.C.: conceptualization, design and methodology, statistical modeling, drafting, and manuscript revision; T.S.: con-

ceptualization, data collection, processing, and manuscript revision; X.B.C.: conceptualization, design, methodology, statistical modeling, drafting, and manuscript revision; A.F.: data collection and processing; J.H.: statistical modeling and manuscript revision; M.K.: conceptualization and manuscript revision; W.C.C.: data collection and manuscript revision; J.B.: conceptualization, methodology, statistical modeling, and manuscript revision; M.K.: conceptualization and manuscript revision; H.I.: data collection and manuscript revision; M.S.: conceptualization and manuscript revision; M.J.K.: data collection, conceptualization, and manuscript revision.

Supplementary data

Supplementary data is available at *ICES Journal of Marine Science* online.

Conflict of interest: The authors declare no conflicts of interest.

Funding

This study was partially funded by the Fisheries Agency, Ministry of Agriculture, Taiwan, through research grant 114AS-6.4.2-FA-01(2), and by the National Science and Technology Council, Taiwan, through research grant 105-2119-M-002 - 058 -MY2 to Yi-Jay Chang.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

References

- Andrews AH, Humphreys RL, Sampaga JD. Blue marlin (*Makaira nigricans*) longevity estimates confirmed with bomb radiocarbon dating. *Can J Fish Aquat Sci* 2018;75:17–25. <https://doi.org/10.1139/cjfas-2017-0031>
- Beamish RJ, Fournier DA. A method for comparing the precision of a set of age determinations. *Can J Fish Aquat Sci* 1981;38:982–3. <https://doi.org/10.1139/f81-132>
- Best N, Cowles MK, Vines K. CODA: convergence diagnosis and output analysis software for Gibbs sampling output version 0.30. MRC Biostatist Unit Cambridge 1995, 41pp.
- Buonaccorsi VP, McDowell JR, Graves JE. Reconciling patterns of inter-ocean molecular variance from four classes of molecular markers in blue marlin (*Makaira nigricans*). *Mol Ecol* 2001;10:1179–96. <https://doi.org/10.1046/j.1365-294x.2001.01270.x>
- Campana SE. Accuracy, precision and quality control in age determination including a review of the use and abuse of age validation methods. *J Fish Biol* 2001;59:197–242. <https://doi.org/10.1111/j.1095-8649.2001.tb00127.x>
- Cayre PM, Diouf T. Estimating age and growth of little tunny, *Euthynnus alletteratus*, off the coast of Senegal, using dorsal fin spines. *NOAA Tech Rep NMFS* 1983;8:105–10.
- Chiang WC, Sun CL, Yeh SZ. Age and growth of sailfin (*Istiophorus platypterus*) in waters off eastern Taiwan. *Fish Bull* 2004;102:251–63.
- Cyr EC, Dean JM, Jehangeer I, *et al.* Age, growth, and reproduction of blue marlin and black marlin from the Indian Ocean. In: R.H. Stroud (Ed.), *Proceedings of the Second International Billfish Symposium Kailua-Kona, Hawaii, August 1–5, 1988, Part 2: National Coalition for Marine Conservation, Inc., Savannah, Georgia, 1990.*
- DeMartini EE, Uchiyama JH, Humphreys RL Jr., *et al.* Age and growth of swordfish (*Xiphias gladius*) caught by the Hawaii-based pelagic longline fishery. *Fish Bull* 2007;105:356–67.
- Dortel E, Massiot-Granier F, Rivot E., *et al.* Accounting for age uncertainty in growth modelling, the case study of yellowfin tuna (*Thunnus albacares*) of the Indian Ocean. *PLoS One* 2013;8:e60886. <https://doi.org/10.1371/journal.pone.0060886>
- Drew K, Die DJ, Arocha F. Understanding vascularization in fin spines of white marlin (*Tetrapturus albidus*). *Bull Mar Sci* 2006;79:847–52.
- Enberg K, Jørgensen C, Dunlop E S., *et al.* Fishing-induced evolution of growth: concepts, mechanisms and the empirical evidence. *Mar Ecol* 2012;33:1–25. <https://doi.org/10.1111/j.1439-0485.2011.00460.x>
- Eveson JP, Million J, Sardenne F., *et al.* Estimating growth of tropical tunas in the Indian Ocean using tag-recapture data and otolith-based age estimates. *Fish Res* 2015;163:58–68. <https://doi.org/10.1016/j.fishres.2014.05.016>
- Farley J, Clear N, Krusic-Golub K., *et al.* A bone to pick with age estimation using hard parts: a case study of swordfish, *Xiphias gladius*, in the southwest Pacific Ocean. *Fish Res* 2022;254:106413. <https://doi.org/10.1016/j.fishres.2022.106413>
- Francis RC. Growth in age-structured stock assessment models. *Fish Res* 2016;180:77–86. <https://doi.org/10.1016/j.fishres.2015.02.018>
- Garcia A, Tserpes G, Santos MN. Validation of annulus formation and growth estimation of South Atlantic swordfish. *J Mar Biol Assoc UK* 2017;97:1511–8. <https://doi.org/10.1017/S0025315416000862>
- Gelman A, Rubin DB. Inference from iterative simulation using multiple sequences. *Statist Sci* 1992;7:457–511. <https://doi.org/10.1214/ss/1177011136>
- Graves JE, McDowell JR. Stock structure of the world's istiophorid billfishes: a genetic perspective. *Mar Freshwater Res* 2003;54:287–98. <https://doi.org/10.1071/MF01290>
- Hatch J, Jiao Y. A comparison between traditional and measurement-error growth models for weakfish *Cynoscion regalis*. *PeerJ* 2016;4:e2431. <https://doi.org/10.7717/peerj.2431>
- Hearn WS, Polacheck T. Estimating long-term growth-rate changes of southern bluefin tuna (*Thunnus maccoyii*) from two periods of tag-return data. *Fish Bull* 2003;101:58–74.
- Heidelberger P, Welch PD. Simulation run length control in the presence of an initial transient. *Oper Res* 1983;31:1109–44. <https://doi.org/10.1287/opre.31.6.1109>
- Hilborn R, Minto-Vera CV. Fisheries-induced changes in growth rates in marine fisheries: are they significant? *Bull Mar Sci* 2008;83:95–105.
- Hill KT, Cailliet GM, Radtke RL. A comparative analysis of growth zones in four calcified structures of Pacific blue marlin, *Makaira nigricans*. *Fish Bull* 1989;87:829–43.
- Hinton MG. Status of blue marlin in the Pacific Ocean. *IATTC Stock Assessment Report* 2001;1:284–319.
- Hoolihan JP, Luo J, Arocha F. Age and growth of blue marlin (*Makaira nigricans*) from the central western Atlantic Ocean. *Fish Res* 2019;220:105346. <https://doi.org/10.1016/j.fishres.2019.105346>
- Ijima H, Shiozaki K. Japanese catch statistics of the Pacific blue marlin (*Makaira nigricans*): update for a stock assessment. *ISC/16/BILLWG-1/02*, 2016, 12pp.
- ISC. Stock assessment update for blue marlin (*Makaira nigricans*) in the Pacific Ocean through 2014. *ISC/16/ANNEX/10*, 2016, 91pp.
- ISC. Stock assessment report for Pacific blue marlin (*Makaira nigricans*) through 2019. *ISC/21/ANNEX/10*, 2021, 115pp.
- Kinney MJ, Chang YJ, Ijima H., *et al.* Length-based proportional sampling for life history research: establishing uniform sampling for North Pacific billfish species. *WCPFC-SC19-2023/SA-IP-11*, 2023, 27pp.
- Kinney MJ, O'Malley J. Collaborative biological sampling of highly migratory species. *ISC/20/BILLWG-01/01*, 2020, 5pp.

- Kleiber P, Hinton MG, Uozumi Y. Stock assessment of blue marlin (*Makaira nigricans*) in the Pacific using MULTIFAN-CL. *Mar Freshwater Res* 2003;54:349–60. <https://doi.org/10.1071/MF01246>
- Kopf RK, Davie PS, Bromhead D., et al. Reproductive biology and spatiotemporal patterns of spawning in striped marlin *Kajikia audax*. *J Fish Biol* 2012;81:1834–58. <https://doi.org/10.1111/j.1095-8649.2012.03394.x>
- Kopf RK, Davie PS, Bromhead D., et al. Age and growth of striped marlin (*Kajikia audax*) in the Southwest Pacific Ocean. *ICES J Mar Sci* 2011;68:1884–95. <https://doi.org/10.1093/icesjms/fsr110>
- Kopf RK, Drew K, Humphreys RL. Age estimation of billfishes (*Kajikia* spp.) using fin spine cross-sections: the need for an international code of practice. *Aquat Liv Res* 2010;23:13–23. <https://doi.org/10.1051/alr/2009045>
- Maunder MN, Punt AE. A review of integrated analysis in fisheries stock assessment. *Fish Res* 2013;142:61–74. <https://doi.org/10.1016/j.fishres.2012.07.025>
- Merrett NR. Gonad development in billfish (Istiophoridae) from the Indian Ocean. *J Zool* 1970;160:355–70. <https://doi.org/10.1111/j.1469-7998.1970.tb03086.x>
- Merrett NR. Aspects of the biology of billfish (Istiophoridae) from the equatorial western Indian Ocean. *J Zool* 1971;163:351–95. <https://doi.org/10.1111/j.1469-7998.1971.tb04539.x>
- Methot RD, Wetzel CR. Stock synthesis: a biological and statistical framework for fish stock assessment and fishery management. *Fish Res* 2013;142:86–99.
- Nakamura I. Billfishes of the world: an annotated and illustrated catalogue of marlins, sailfishes, spearfishes and swordfishes known to date. *FAO Species Catalogue*, 1985. vol. 5 FAO Fish Synop. 125:66pp.
- Nishikawa Y, Honma M, Ueyanagi S, Kikawa S. Average distribution of larvae of oceanic species of scombrid fishes, 1956–1981. *Far Seas Fish Res Lab S Ser* 1985;12:99.
- Ogle DH, Wheeler P, Dinno A. FSA: Fisheries Stock Analysis. R package version 0.8.30, 2020. Available at: <https://github.com/droglenc/FSA> (12 June 2024, date last accessed).
- Orbesen ES, Hoolihan JP, Serafy JE., et al. Transboundary movement of Atlantic istiophorid billfishes among international and U.S. Domestic management areas inferred from mark-recapture studies. *Mar Fish Rev* 2008;70:14–23.
- Pons M, Arocha F, Domingo A., et al. Potential use of anal fin spines for assessing age and growth of longbill spearfish, *Tetrapturus pfluegeri*, in the western Atlantic Ocean. *Bull Mar Sci* 2016;92:385–98. <https://doi.org/10.5343/bms.2015.1043>
- Prince ED, Lee DW, Berkeley SA. Use of marginal increment analysis to validate the anal spine method for ageing Atlantic swordfish and other alternatives for age determination. *Col Vol Sci Pap ICCAT* 1988;27:194–201.
- Prince ED, Lee DW, Zwiefel JR., et al. Estimating age and growth of young Atlantic blue marlin *Makaira nigricans* from otolith microstructure. *Fish. Bull.* 1991;89:441–59.
- R Core Team. *R: a Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2018. Available at: <https://www.R-project.org/> (29 June 2024, date last accessed)
- Schneider CA, Rasband WS, Eliceiri KW. NIH image to ImageJ: 25 years of image analysis. *Nat Methods* 2012;9:671–5. <https://doi.org/10.1038/nmeth.2089>
- Schnute J. A versatile growth model with statistically stable parameters. *Can J Fish Aquat Sci* 1981;38:1128–40. <https://doi.org/10.1139/f81-153>
- Shimose T, Fujita M, Yokawa K., et al. Reproductive biology of the blue marlin *Makaira nigricans* around Yonaguni Island, southwestern Japan. *Fish Sci* 2009;75:109–19. <https://doi.org/10.1007/s12562-008-0006-8>
- Shimose T, Yokawa K, Tachihara K. Age determination and growth estimation from otolith micro-increments and fin spine sections of blue marlin (*Makaira nigricans*) in the western North Pacific. *Mar Freshw Res* 2015;66:1116–27. <https://doi.org/10.1071/MF14305>
- Sorenson L, McDowell JR, Knott T., et al. Assignment test method using hypervariable markers for blue marlin (*Makaira nigricans*) stock identification. *Conserv. Genet Res* 2013;5:293–7. <https://doi.org/10.1007/s12686-012-9747-x>
- Speare P. Age and growth of black marlin, *Makaira indica*, in east coast Australian waters. *Mar Freshwater Res* 2003;54:307–14. <https://doi.org/10.1071/MF01277>
- Spiegelhalter DJ, Best NG, Carlin BR., et al. Bayesian measures of model complexity and fit. *J R Statist Soc Ser B Statist Methodol* 2002;64:583–639. <https://doi.org/10.1111/1467-9868.00353>
- Spiegelhalter DJ, Thomas A, Best NG., et al. *WinBUGS Version 1.4 User Manual*. Cambridge: MRC Biostatistics Unit, 2007.
- Sun CL, Chang YJ, Tzeng CC., et al. Reproductive biology of blue marlin (*Makaira nigricans*) in the western Pacific Ocean. *Fish Bull* 2009;107:420–32.
- Sun CL, Wang SP, Yeh SZ. Age and growth of the swordfish (*Xiphias gladius*) in the waters around Taiwan determined from anal fin rays. *Fish Bull* 2002;100:822–35.
- Sun CL, Yeh SZ, Liu CS., et al. Age and growth of black marlin (*Istiompax indica*) off eastern Taiwan. *Fish Res* 2015;166:4–11. <https://doi.org/10.1016/j.fishres.2014.09.005>
- Williams SM, Wyatt J, Ovenden JR. Investigating the genetic stock structure of blue marlin (*Makaira nigricans*) in the Pacific Ocean. *Fish Res* 2020;228:105565. <https://doi.org/10.1016/j.fishres.2020.105565>
- Wilson CA, Dean JM, Prince ED., et al. An examination of sexual dimorphism in Atlantic and Pacific blue marlin using body weight, sagittae weight, and age estimates. *J Exp Mar Biol Ecol* 1991;151:209–25. [https://doi.org/10.1016/0022-0981\(91\)90125-G](https://doi.org/10.1016/0022-0981(91)90125-G)
- Zar JH. *Biostatistical Analysis*. 4th Edition, Prentice Hall, Upper Saddle River, 1999.

Handling editor: Allen Andrews